

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. 3rd Semester Mid-Term Examination, 2023
Subject: Digital Logic (CS - 2102)

Full marks: 30

Time: 2 hours

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1. Minimize the following logic expressions.
 - (a) $f(x,y,z) = xy + x'z + yz$
 - (b) $f(x,y,z) = x'yz + xy'z + xyz' + xyz$ [2.5 x 2 = 5]
 2. (a) Draw the minimized logic circuit for the following Boolean expression using NAND gates only.
 $f(x,y) = x' + xy$.
(b) Draw a logic circuit to realize AND operation using NOR gates. [2.5x2=5]
 3. Define encoder. What is the basic difference between encoder and priority encoder? Design a four input priority encoder using two input NAND gate. [1+1+3=5]
 4. Perform the arithmetic operations in binary using the signed-2's-complement representation for negative numbers.
 - (a) $(+32) + (-23)$
 - (b) $(-32) - (-23)$ [2.5 x 2 = 5]
 5. Design full subtractor using 4:1 MUX.
Write the truth table of full subtractor
Show the design details
Draw the logic circuit [2+2+1=5]
 6. Design BCD to excess-3 code converter
Write the truth table of code converter
Show the design details
Draw the logic circuit [2+2+1=5]